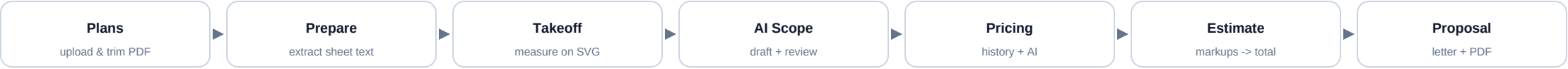


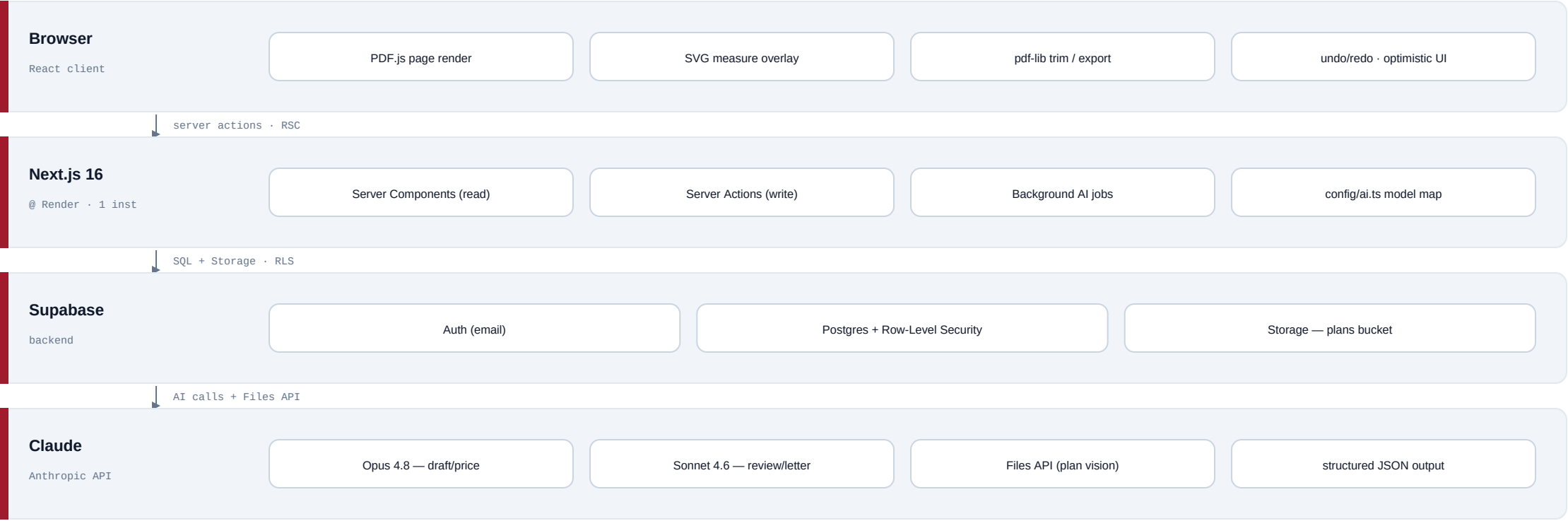
XtraUnit Estimator — Architecture & Data Flow

Next.js 16 (App Router) · React 19 · Supabase (Postgres + RLS) · Anthropic Claude · pdf.js / pdf-lib · Render

1 · PRODUCT PIPELINE (plans -> proposal)



2 · ARCHITECTURE (a request flows top -> bottom; data returns bottom -> top)



Signature pattern — background AI jobs (scope & pricing)

Known limit: single instance – a job queue + worker is the next infra step.

Server Action inserts a scope_runs row -> spawns a fire-and-forget job in the SAME process -> job re-auths with the user's token -> gatherBundle -> Files API (streamed 1 file/time) -> draftScope (Opus, chunked) -> findGaps (Sonnet) -> writes line_items + scope_findings; UI polls progress.

XtraUnit Estimator — Database Schema (Supabase / Postgres)

15 tables · every table is owner-scoped via Row-Level Security (owner_id -> auth.users). •=primary key ->=foreign key reference uniq=unique

projects	
• id	uuid
owner_id	-> auth.users
name	text
client_name	text
address	text
project_type	text
status	text
notes	text
region	text
created_at	ts
updated_at	ts

plan_files	
• id	uuid
project_id	-> projects
owner_id	-> auth.users
file_name	text
storage_path	text
size_bytes	int8
mime_type	text
created_at	ts

sheets	
• id	uuid
project_id	-> projects
plan_file_id	-> plan_files
owner_id	-> auth.users
page_number	int
original_page_number	int
name	text
label	text
notes	text
scale_px_per_unit	num
scale_unit	text
scale_x	num
scale_y	num
scale_preset	text
extracted_text	text
ingest_method	text
ingested_at	ts
ledger	jsonb
created_at	ts

profiles	
• id	-> auth.users
email	text
full_name	text
role	text
created_at	ts

measurements	
• id	uuid
project_id	-> projects
plan_file_id	-> plan_files
sheet_id	-> sheets
owner_id	-> auth.users
type	text
geometry	jsonb
value	num
unit	text
layer	text
color	text
attributes	jsonb
wall_sided	text
wall_height	num
vol_mode	text
vol_width	num
vol_depth	num
text	text
font_size	num
head_size	num
created_at	ts

company_settings	
• id	uuid
owner_id	-> auth.users · uniq
company_name	text
company_address	text
company_phone	text
company_email	text
company_license	text
default_contingency_pct	num
default_insurance_pct	num
default_op_pct	num
signer_name	text
signer_title	text
benchmarks	jsonb
unit_prices	jsonb
proposal_profile	jsonb
created_at	ts
updated_at	ts

line_items	
• id	uuid
project_id	-> projects
plan_file_id	-> plan_files
owner_id	-> auth.users
division_code	text
division_name	text
section_code	text
section_name	text
description	text
quantity	num
unit	text
source_kind	text
evidence	jsonb
status	text
confidence	text
ai_generated	bool
user_edited	bool
notes	text
sort_order	int
price_mode	text
cost_labor	num
cost_material	num
cost_sub	num
cost_equipment	num
cost_other	num
cost_total	num
price_source	text
price_note	text
price_confidence	text
priced_at	ts
sub_quote_id	-> sub_quotes
created_at	ts
updated_at	ts

scope_runs	
• id	uuid
project_id	-> projects
owner_id	-> auth.users
status	text
stage	text
progress	int
error	text
kind	text
created_at	ts
updated_at	ts

cost_database	
• id	uuid
owner_id	-> auth.users
project_id	-> projects
item_id	-> cost_items
division_code	text
section_code	text
description	text
unit	text
price_mode	text
cost_labor	num
cost_material	num
cost_sub	num
cost_equipment	num
cost_other	num
cost_total	num
price_source	text
price_note	text
price_confidence	text
source	text
region	text
project_type	text
building_sf	num
observed_on	date
created_at	ts

cost_items	
• id	uuid
owner_id	-> auth.users
division_code	text
section_code	text
name	text
norm_key	text
unit	text
aliases	jsonb
std_cost_override	num
std_cost_computed	num
std_count	int
last_observed	ts
active	bool
created_at	ts
updated_at	ts

scope_findings	
• id	uuid
project_id	-> projects
plan_file_id	-> plan_files
owner_id	-> auth.users
kind	text
text	text
severity	text
evidence	jsonb
resolved	bool
answer	text
answered_at	ts
created_at	ts

sub_quotes	
• id	uuid
project_id	-> projects
owner_id	-> auth.users
sub_name	text
trade	text
division_codes	text[]
quote_date	text
total	num
file_path	text
file_name	text
extracted	jsonb
notes	text
created_at	ts

estimates	
• id	uuid
project_id	-> projects · uniq
owner_id	-> auth.users
contingency_pct	num
insurance_pct	num
overhead_pct	num
profit_pct	num
building_sf	num
notes	text
created_at	ts
updated_at	ts

proposals	
• id	uuid
project_id	-> projects · uniq
owner_id	-> auth.users
letter_text	text
client_name	text
proposal_date	text
project_description	text
understanding	text
estimated_duration	text
anticipated_start	text
table_style	text
created_at	ts
updated_at	ts